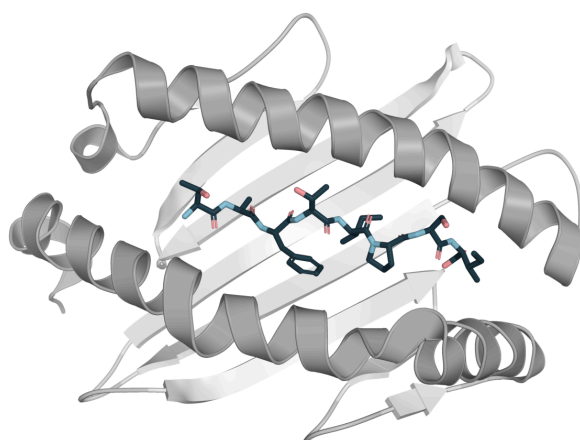


## 1 Design challenge

The computational prediction of protein-protein interaction stability is a key step in protein design workflows. An example of this is modern immunotherapy design, which requires the prediction of peptide-HLA complex stability. Many teams use NetMHCstabpan<sup>1</sup>, a sequence-based, domain-specific supervised learning model trained on peptide-HLA sequence pairs. This challenge asks if we can use existing protein foundation models to better predict the stability of peptide-HLA complexes.

## 2 Biological context

The immune system is able to distinguish between healthy protein sequences and foreign, potentially pathogenic protein sequences through a variety of mechanisms. One of these mechanisms involves the continued processing and presentation of protein fragments (or ‘peptides’) on the cell surface for surveillance by circulating T cells. This presentation of fragments is facilitated by a highly conserved class of cell surface protein called human leukocyte antigens or HLAs (Figure 1). This task focuses on class I HLAs, which bind peptides of typically 8-11 amino acids, and on the resulting peptide-HLA complex stability. While highly conserved, different HLA alleles bind distinct groups of peptides; each individual will present different sets of peptides on their cell surfaces dependent on their HLA genotype. Hence, the computational ranking of which peptides will stably bind a given HLA allele is an important problem in the design of effective personalised immunotherapies.



**Figure 1** A peptide bound in a class I groove. The HLA alpha1/alpha2 domain is shown in grey; the peptide is shown as sticks, carbon in dark blue, oxygen coral, nitrogen pale blue. Two alpha helical regions of the HLA form a conserved groove where the peptide binds. Rendered from PDB 4MJ1<sup>2</sup>.

This task focuses on peptide-HLA complex stability, which is related to but distinct from binding affinity. While binding affinity is a function of both association rate (or “on rate”) and dissociation rate (or “off rate”), stability measures how long a peptide stays in the binding groove once bound, and is largely a function of the dissociation rate.

## 3 Protein foundation models

In recent years, open source protein foundation models have become the cornerstone of modern protein design workflows. For this task, we focus broadly on three classes of model:

**Structure prediction models:** These models take a protein sequence and predict the coordinates of the 3D protein structure.

**Inverse folding models:** These models do the inverse of structure prediction; given a set of 3D coordinates, they predict protein sequences which will adopt those coordinates.

**Protein language models:** These models take a protein sequence and represent it in an embedding space which can be used for a variety of downstream tasks; these representations contain rich co-evolutionary information and sequence context.

| Foundation model | Paper                                                                                                                             | GitHub                                                                                                                                                              | Model type             |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Boltz-2          | <a href="#">Boltz-2: Towards Accurate and Efficient Binding Affinity Prediction</a> (bioRxiv, 2025)                               | <a href="https://github.com/jwohlwend/boltz">https://github.com/jwohlwend/boltz</a>                                                                                 | Structure prediction   |
| Chai-1           | <a href="#">Chai-1: Decoding the molecular interactions of life</a> (bioRxiv, 2024)                                               | <a href="https://github.com/chaidisco/very/chai-lab">https://github.com/chaidisco/very/chai-lab</a>                                                                 | Structure prediction   |
| Protenix         | <a href="#">Protenix - Advancing Structure Prediction Through a Comprehensive AlphaFold3 Reproduction</a> (bioRxiv, 2025)         | <a href="https://github.com/bytedance/Protenix">https://github.com/bytedance/Protenix</a>                                                                           | Structure prediction   |
| ESMFold          | <a href="#">Evolutionary-scale prediction of atomic-level protein structure with a language model</a> ( <i>Science</i> , 2023)    | <a href="https://github.com/facebookresearch/esm">https://github.com/facebookresearch/esm</a>                                                                       | Structure prediction   |
| ESMFold2         | <a href="#">Language Modeling Materializes a World Model of Protein Biology</a> (bioRxiv, 2026)                                   | <a href="https://github.com/Biohub/esm">https://github.com/Biohub/esm</a>                                                                                           | Structure prediction   |
| ProteinMPNN      | <a href="#">Robust deep learning-based protein sequence design using ProteinMPNN</a> ( <i>Science</i> , 2022)                     | <a href="https://github.com/dauparas/ProteinMPNN">https://github.com/dauparas/ProteinMPNN</a>                                                                       | Inverse folding        |
| LigandMPNN       | <a href="#">Atomic context-conditioned protein sequence design using LigandMPNN</a> ( <i>Nature Methods</i> , 2025)               | <a href="https://github.com/dauparas/LigandMPNN">https://github.com/dauparas/LigandMPNN</a>                                                                         | Inverse folding        |
| ESM-IF           | <a href="#">Learning inverse folding from millions of predicted structures</a> ( <i>ICML</i> , 2022)                              | <a href="https://github.com/facebookresearch/esm/tree/main/examples/inverse_folding">https://github.com/facebookresearch/esm/tree/main/examples/inverse_folding</a> | Inverse folding        |
| ESM-2            | <a href="#">Evolutionary-scale prediction of atomic-level protein structure with a language model</a> ( <i>Science</i> , 2023)    | <a href="https://github.com/facebookresearch/esm">https://github.com/facebookresearch/esm</a>                                                                       | Protein language model |
| ESMC             | <a href="#">Language Modeling Materializes a World Model of Protein Biology</a> (bioRxiv, 2026)                                   | <a href="https://github.com/Biohub/esm">https://github.com/Biohub/esm</a>                                                                                           | Protein language model |
| ProtT5           | <a href="#">ProtTrans: Toward Understanding the Language of Life Through Self-Supervised Learning</a> ( <i>IEEE TPAMI</i> , 2022) | <a href="https://github.com/agemagic/ProtTrans">https://github.com/agemagic/ProtTrans</a>                                                                           | Protein language model |
| SaProt           | <a href="#">SaProt: Protein Language Modeling with Structure-aware Vocabulary</a> ( <i>ICLR</i> , 2024 spotlight)                 | <a href="https://github.com/westlake-repl/SaProt">https://github.com/westlake-repl/SaProt</a>                                                                       | Protein language model |

#### 4 Dataset for this challenge

The dataset for this challenge is the NetMHCstabpan<sup>1</sup> peptide-HLA stability dataset. The data are supplied as a single CSV file containing 28,166 peptide-HLA half-life measurements with the following columns:

| Column        | Description                                                                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allele        | HLA class I allele in IPD-IMGT/HLA nomenclature <sup>3</sup> , e.g. <i>HLA-A*02:01</i> . Three engineered constructs retain a suffix denoting their substitution, e.g. <i>HLA-B*14:01(C67S)</i> . |
| peptide       | Peptide sequence in one-letter amino acid code. All peptides are 9-mers.                                                                                                                          |
| thalf_hours   | Measured half-life of the peptide-HLA complex, in hours. This is the prediction target.                                                                                                           |
| hla_seq       | The HLA alpha1/alpha2 domain sequence, which forms the peptide-binding groove.                                                                                                                    |
| hla_pseudoseq | The 34-residue pseudosequence of peptide-contacting positions used by the NetMHCpan family of methods <sup>4</sup> .                                                                              |

We gratefully acknowledge Rasmussen et al.<sup>1</sup> for generating this dataset and making it publicly available; submissions making use of it should cite their work.

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## 5 Notes on the challenge

We are not looking for an approach that tops an arbitrary leaderboard. NetMHCstabpan<sup>1</sup>, a competitive baseline for this task, is trained on the entirety of this dataset, hence is unfair to use as a direct comparator. We anticipate great teams using this data to answer the research question above; can existing foundation models improve performance of peptide-HLA stability prediction? To this end, we would recommend keeping a few things in mind:

- We provide no splits in this dataset specifically to avoid a leaderboard approach; we would strongly recommend establishing splits early on with a sensible hypothesis.
  - A simple supervised neural network trained on peptide and HLA pairs would be a helpful baseline to produce to contextualise the results on your splits.
  - Different types of open foundation models will require different levels of compute and runtime for inference; this should be taken into account either through data subsetting or careful choice of which model(s) to evaluate.
  - These models can be used in multiple ways - they produce embeddings, log-likelihoods/perplexities, and both internal and external confidence metrics. They can be run with different inputs masked and different seeds, fine-tuned or have prediction heads built on top of their embeddings. We encourage you to explore under the hood!
  - Negative results are just as good as positive results where well supported. Foundation models are very capable, but computationally heavy and expensive to produce and run. We are asking the question *Are they useful for this problem*, not *Please prove they are useful for this problem*.
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## 6 Evaluation criteria

We're interested in how you tackle this question, your approach to the data, chosen evaluation metrics, your conclusions, and the evidence you provide to support them. Winning submissions will have undertaken a watertight evaluation considering ML best practices, engineering & compute requirements during the timeframe, and incorporation of the biological background of the problem. This is purposefully designed as an open question with a lot of degrees of freedom in your approach; we're excited to see your submissions!

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## 7 Citations

- 1 Rasmussen M, Fenoy E, Harndahl M, Kristensen AB, Nielsen IK, Nielsen M, Buus S. Pan-Specific Prediction of Peptide-MHC Class I Complex Stability, a Correlate of T Cell Immunogenicity. *The Journal of Immunology*. 2016;197(4):1517–1524. [doi:10.4049/jimmunol.1600582](https://doi.org/10.4049/jimmunol.1600582)
  - 2 Motozono C, Kuse N, Sun X, Rizkallah PJ, Fuller A, Oka S, Cole DK, Sewell AK, Takiguchi M. Molecular Basis of a Dominant T Cell Response to an HIV Reverse Transcriptase 8-mer Epitope Presented by the Protective Allele HLA-B\*51:01. *The Journal of Immunology*. 2014;192(7):3428–3434. [doi:10.4049/jimmunol.1302667](https://doi.org/10.4049/jimmunol.1302667). Structure shown in Figure 1 rendered from PDB 4MJI.
  - 3 Barker DJ, Maccari G, Georgiou X, Cooper MA, Flicek P, Robinson J, Marsh SGE. The IPD-IMGT/HLA Database. *Nucleic Acids Research*. 2023;51(D1):D1053–D1060. [doi:10.1093/nar/gkac1011](https://doi.org/10.1093/nar/gkac1011)
  - 4 Reynisson B, Alvarez B, Paul S, Peters B, Nielsen M. NetMHCpan-4.1 and NetMHCIIpan-4.0: improved predictions of MHC antigen presentation by concurrent motif deconvolution and integration of MS MHC eluted ligand data. *Nucleic Acids Research*. 2020;48(W1):W449–W454. [doi:10.1093/nar/gkaa379](https://doi.org/10.1093/nar/gkaa379)
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